

A Comparative Study of Symmetric Cryptography Mechanism on DES, AES and EB64 for Information Security

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Abstract- The usages of Information technology were numerous in every aspect of life. As it were, an adequate security has often been a great challenge which has to be more focused on. Different methods have been employed to secure and protect the shared and sensitive data. This research work basically focused on cryptography in ensuring zero tolerance for insecurity of data in transit. However, in order to improve security and efficiency of valuable data, the Private or Secret Key Encryption Infrastructure needs to be adopted as the model to implement security. This research work implemented three encryption techniques such as DES, AES and EB64 algorithms and compared their performances on both encryption and decryption techniques based on the analysis of encryption and decryption time allotted at each different experimental stage. The experimental results obtained were used to analyse the effectiveness of each algorithm on a mobile phone. The result that was gathered during the various experiments for SMS plaintext shows that EB64 encryption and decryption time was shorter as compared with AES and DES.

Keywords: Cryptography, Encryption, Decryption, Private key encryption, Security, AES, DES, EB64

I. INTRODUCTION

In the current world, encryption plays a significant role in securing relevant data or information from eavesdroppers, attackers, and unauthorized users. A lot of attacks occurred when an unknown user tries to intercept communication between two authorized devices. Therefore, data security is very crucial in maintaining the confidentiality of information especially containing sensitive information that should only be known by certain parties. However, many encryption algorithms are widely available and used in information security. These can be grouped into two namely; Symmetric (private) and Asymmetric (public) keys encryption. In Symmetric keys encryption or secret key encryption, only one key is used to encrypt and decrypt data. In Asymmetric keys, two keys are used; private and public keys [11]. Public key is used for encryption and private key is used for decryption (e.g. RSA). Public key encryption is based on mathematical functions, computationally intensive. There are many examples of encryption algorithms such as DES, AES, EB64 and RSA etc. Asymmetric key encryption or public key encryption is used to solve the problem of key distribution. In Asymmetric keys, two keys are used; private and public keys. Public key is used for encryption and private key is used for decryption (E.g. RSA and Digital Signatures). Because users tend to use two keys: public key, which is known to the public and

private key which is known only to the user [2]. There is no need for distributing them prior to transmission. However, public key encryption is based on mathematical functions, computationally intensive and is not very efficient for small mobile devices [5]. Asymmetric encryption techniques are almost 1000 times slower than Symmetric techniques, because they require more computational processing power [11]. This study evaluates three different encryption algorithms namely; AES, DES and EB64. The performance measure of encryption schemes is conducted in terms of encryption and decryption time using several SMS messages.

The rest of the paper is arranged as follows: section II provides a background of the existing literature and some related works were presented. Section III focuses on the methodology by presenting a comparative study of the three aforementioned symmetric algorithms. Section IV demonstrates the outcome of comparative study in III and analysis of the various experimental results were done. Section V is conclusion; Section VI is contribution to knowledge, and Section VII is future work.

II. LITERATURE REVIEW

A. A Conceptual review of Encryption Algorithms

According to [9], defines encryption as a well known technology for protecting sensitive data. In the review, the algorithm used the combination of Public and Private Key

encryption to hide the sensitive data of users, and cipher text retrieval.

Overview of Data Security Algorithm

❖ Data Encryption Standard (DES)

DES is a block cipher that uses shared secret key for encryption and decryption. DES algorithm as described by Amalraj [3] takes a fixed length of string in plaintext bits and transforms it through a series of operations into cipher text bit string of the same length and its each block is 64 bits. There are 16 identical stages of processing, termed rounds. There is also an initial and final permutation which named as IP and FP respectively.

❖ Advanced Encryption Standard (AES)

In AES [6] is the almost identical of block cipher Rijndael cipher developed by two Belgian cryptographers, Joan and Vincent Rijmen. The algorithm explains about by AES is a secret-key algorithm which means of the same key is used for both encrypting and decrypting the data.

AES on the other hand which encrypts all 128 bits in one iteration. This is one reason why it has a comparably small number of rounds. AES encryption is fast and flexible. It can be implemented on various platforms especially in small devices.

❖ Enhanced Base64 Algorithm (EB64)

An enhanced base64 algorithm (EB64) is an algorithm developed to encode and decode large amount of textual data using a private key [15]. The scheme uses a concept of modern encryption algorithms whereby a key is XOR-ed into the existing Base64 algorithmic process to enhance the security features of the algorithm. This enhanced algorithm is a secret-key algorithm which means of the same key is used for both encrypting and decrypting the data.

B Review of Related works

Monther et, al, [13] authored real implementation for SMS encryption-based on android message. The study developed an SMS encryption for mobile communication on android message application because of its insecurity during transmission. The study further implemented three of block cipher symmetric cryptography algorithms (i.e. AES algorithm, DES, and 3-DES) and compared three of them in terms of encryption and decryption delay time in order to know the most suitable cryptography algorithm for mobile communication on android message application. Thus, the experiment tests shows the DES encryption algorithm has low encryption delay time in different message sizes (i.e. 32, 64, 128, 256, 512, and 1024) bits.

Logunleko et, al., [15] did a research on enhancement of Base64 algorithm as Base64 algorithm was not adequately secure as a result of unavailability of key. The paper introduced a key mechanism into the algorithmic flow of the existing base64 algorithm to increase the security capability of the algorithm as compared with the non-key based algorithm.

Pavithra et, al., [14] presented a study on the performance evaluation of cryptographic algorithms. The authored compared the various cryptographic algorithms such as AES, DES and Blowfish on the basis of time as parameter. These various cryptographic algorithms are evaluated on different video files. Different video files are having different processing speed on which various size of file are processed. The authors observed that AES algorithm is executed lesser processing time and more throughput level as compared to the other algorithms. The study then proposed that evaluation of the performance of audio and video files for other parameters such as, memory usage and output byte in the future.

Shashi et, al, [16]. did a comparative analysis of encryption algorithms for data communication on AES, DES and RSA using cryptographic tools for conducting experiments to show the effectiveness of each algorithm on the same text files based on three parameters such as computation size, memory usage and outputs byte respectively. In the study, RSA takes larger memory than AES and DES but RSA algorithm produces low level of output byte. The author further said that the future work will be focused on image and audio data as well as to improve encryption time and less memory.

Amalraj and Raybin [3] did a review on study of cryptography techniques such as AES, DES, 3DES, Blowfish, RSA etc. Among these algorithms, the security for the data has become highly important since the selling and buying of products over the open network occur very frequently. The authors surveyed the existing works on the encryption techniques and presented the performance evaluation of selected symmetric algorithms. It was therefore concluded that Blowfish has the better performance than other algorithms. For future work, encryption techniques could be used in such a way that it could consume less time and power, minimum energy consumption and high speed.

Paritosh Patil [10] proposed encryption for secure SMS transmission. In this review, an enhanced security technique was adopted using an advanced version of AES called 3D-AES which generates a symmetric key by shuffling the original key array three times and taking the key better each time it is shuffled. The 3D-AES is to improve confusion performance of round transformation. In addition, PGP (Pretty good privacy) technique was also adopted which provides cryptographic privacy and authentication for data communication. Therefore, an end-to-end key based encryption technology for SMS plugs the gaps of insecurity in SMS transmission. Hence, SMS users can now exchange sensitive information via SMS through the proposed technique which prevents attackers to intercept the transmission.

Agoyi et. al., [2] revealed a framework that could be used basically for mobile communication. The framework provides a solution to SMS security problem. The approach used for the work was to secure the SMS

message using Hybrid Compression Encryption (HCE) system. This system compresses the SMS to reduce its length, then encrypts it using RSA algorithm. RSA is a Public Key Encryption method. A disadvantage of using public-key cryptography for encryption is speed.

III. METHODOLOGY

A comparative study between DES, AES and EB64 was presented by considering nine (9) factors which are Key Size, Block Size, Scalability, Algorithm, Encryption, Decryption, Power Consumption, Security, Key Used, Rounds, Hardware and Software Implementation. The developed comparison model was then implemented using

sublime text, Javascript, HTML5, mobile phone and Phonegap technology with minimum of 2.0 GHz processor size, 2 GB of RAM and 16GB of storage Space. The performance evaluation of the three mentioned algorithms was carried out using encryption and decryption time as parameter.

IV. DISCUSSION OF RESULTS

A. Comparison

Based on the nine (9) factors aforementioned, the comparisons analyses of DES, AES and EB64 were tabulated in table 1 below.

Table 1: Comparison between DES, AES and EB64.

Factors	DES	AES	EB64
Key Size	56bits	128,192,256 bits	Greater than or equal to 1
Block Size	64 bits	128 bits	Unlimited
Scalability	It is a Scalable algorithm due to varying the key size and Block size	Not Scalable	Scalable
Algorithm	Symmetric	Symmetric	Symmetric
Encryption	Moderate	Faster	Faster than both
Decryption	Moderate	Faster	Faster than both
Power Consumption	Low	Low	Low
Security	Not secured enough	Excellent Secured	Adequately secured
Key Used	Same key used for Encrypt and Decrypt	Same key used for Encrypt and Decrypt	Same key used for Encrypt and Decrypt

B. Experiment Result Analysis

The developed model was carried out among these three (3) algorithms DES, AES and EB64 respectively. The study conducted five (5) different experiments each to analyse and evaluate the performance of each of the algorithm.

In stage 1, as shown in table 2. The analysis and evaluation of result shows that the average of the EB64 algorithm in respect to DES and AES for a size of 15 Bytes data during encryption and decryption time. EB64 encryption and decryption was 0.002 seconds and 0.001 seconds while DES and AES encryption and decryption were 0.009 seconds and 0.008 seconds, 0.008 seconds and 0.004 seconds respectively.

Stage 2, in another related experiment shown in table 3 with SMS size of 24Bytes. The encryption and decryption EB64 was 0.001 seconds and 0.002 seconds while DES and AES encryption and decryption were 0.011 seconds and 0.006 seconds, 0.009 seconds and 0.005 seconds respectively.

Another experiment was performed in stage 3 with SMS size 41 bytes as shown in table 4, encryption and decryption time for EB64 was 0.002 seconds and 0.003 seconds, AES was 0.011 seconds and 0.005 seconds and DES was 0.012 and 0.015 seconds respectively.

Similarly, the result gathered during the experiment for SMS plaintext 67 bytes as depicted in table 5 shows that encryption time was 0.003 seconds and decryption time was 0.004 seconds for EB64 as compared with AES encryption time of 0.016 seconds and decryption time of 0.01 seconds while DES with encryption time of 0.009 seconds and decryption time of 0.005 seconds respectively.

In the last stage, the experiment for SMS plaintext 103 bytes as shown in table 6, EB64 encryption and decryption time was 0.005 seconds and 0.007 seconds while AES encryption and decryption time was 0.012 seconds and 0.008 seconds, and DES encryption and decryption time was 0.018 seconds and 0.012 seconds.

Table 2: Analysis of SMS Plaintext Size - 15Bytes

Algorithm	DES		AES		EB64	
S/No	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)
1	0.023	0.011	0.018	0.006	0.002	0.003
2	0.007	0.009	0.008	0.004	0.002	0.001

3	0.004	0.006	0.004	0.005	0.001	0.001
4	0.005	0.008	0.005	0.003	0.001	0.001
5	0.004	0.004	0.004	0.004	0.002	0.001
AVG_i	0.009	0.008	0.008	0.004	0.002	0.001

Table 3: Analysis of SMS Plaintext Size - 24Bytes

Algorithm						
	DES		AES		EB64	
S/No	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (sec)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)
1	0.022	0.011	0.022	0.007	0.001	0.003
2	0.006	0.006	0.008	0.005	0.001	0.002
3	0.012	0.005	0.006	0.005	0.002	0.002
4	0.005	0.005	0.005	0.004	0.002	0.001
5	0.009	0.004	0.005	0.003	0.001	0.001
AVG_{ii}	0.011	0.006	0.009	0.005	0.001	0.002

Table 4: Analysis of SMS Plaintext Size - 41Bytes

Algorithm						
	DES		AES		EB64	
S/No	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)
1	0.033	0.019	0.032	0.007	0.003	0.003
2	0.007	0.013	0.005	0.006	0.001	0.003
3	0.006	0.007	0.006	0.004	0.002	0.002
4	0.007	0.005	0.006	0.004	0.001	0.003
5	0.005	0.029	0.005	0.004	0.005	0.002
AVG_{iii}	0.012	0.015	0.011	0.005	0.002	0.003

Table 5: Analysis of SMS Plaintext Size - 67Bytes

Algorithm						
	DES		AES		EB64	
S/No	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)
1	0.026	0.018	0.022	0.007	0.004	0.005
2	0.012	0.009	0.006	0.006	0.002	0.004
3	0.008	0.01	0.006	0.005	0.003	0.004
4	0.007	0.007	0.006	0.005	0.002	0.003
5	0.027	0.007	0.007	0.004	0.003	0.003
AVG_{iv}	0.016	0.01	0.009	0.005	0.003	0.004

Table 6: Analysis of SMS Plaintext Size - 103Bytes

Algorithm						
	DES		AES		EB64	
S/No	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)	Encryption Time (secs)	Decryption Time (secs)
1	0.039	0.015	0.032	0.011	0.013	0.01
2	0.013	0.011	0.007	0.008	0.003	0.006
3	0.013	0.012	0.007	0.007	0.003	0.005
4	0.014	0.01	0.008	0.007	0.004	0.006
5	0.011	0.013	0.007	0.005	0.004	0.006
AVG_v	0.018	0.012	0.012	0.008	0.005	0.007

Furthermore, the experimental results for Encryption algorithm AES, DES and EB64 were shown in Table 2 - Table 6, which shows the comparison of the three algorithms using same SMS messages size for each five different experiments. The average of each same size of encryption and decryption time was then calculated.

The calculated average of each encryption and decryption time of the three Algorithms in table 2, 3, 4, 5 and 6, were

recorded in table 7, which was then used to plot an encryption and decryption time graph in order to show the disparities of the relationship between the three Algorithms. Hence, table 7 shows the five (5) different averages of DES, AES and B64 Encryption and Decryption time using the same plain text sizes as earlier said.

Table 7: Shows average of each encryption and decryption time of the three Algorithms in different stages of the experiments.

S/NO	Algorithm	Plaintext size in Bytes	Encryption Time (Secs)	Decryption Time (Secs)
AVGi	DES	15	0.009	0.008
	AES		0.008	0.004
	EB64		0.002	0.001
AVGii	DES	24	0.011	0.006
	AES		0.009	0.005
	EB64		0.001	0.002
AVGiii	DES	41	0.012	0.015
	AES		0.011	0.005
	EB64		0.002	0.003
AVGiv	DES	67	0.016	0.01
	AES		0.009	0.005
	EB64		0.003	0.004
AVGv	DES	103	0.018	0.012
	AES		0.012	0.008
	EB64		0.005	0.007

By analyzing table 7, time taken by EB64 algorithm (AVGi - AVGv) for both encryption and decryption

process is lesser compare to the time taken by AES and DES algorithm.

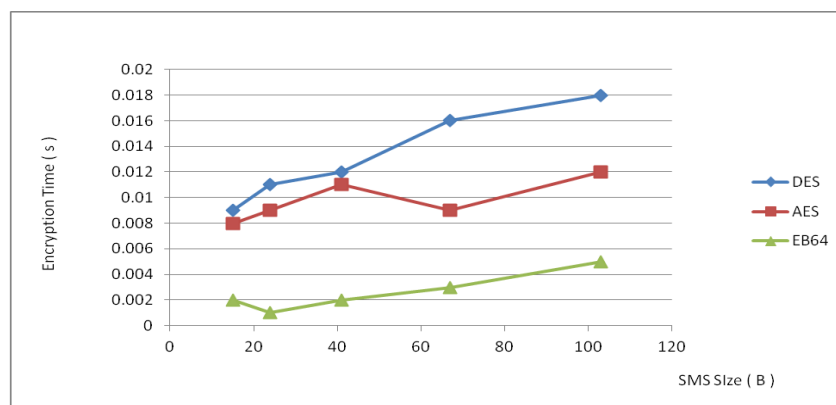


Figure 1: Comparison of Encryption Time among DES, AES and EB64

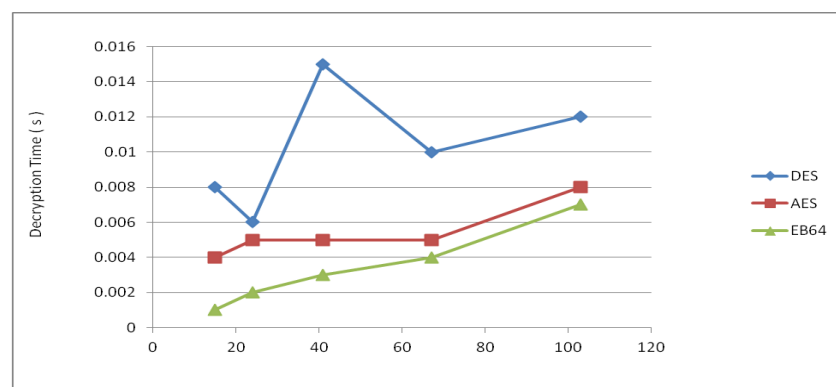


Figure 2: Comparison of Decryption Time among DES, AES and EB64

Thus, by analyzing figure 1 and figure 2 which show the time taken for encryption and decryption on various sizes of SMS by the three algorithms. EB64 algorithm takes much shorter time compare to time taken by DES and AES algorithm.

V. CONCLUSION

The significant roles of encryption algorithms are numerous and essential in information security. However, this research work carried out the performance evaluation of the enhanced Base64 Algorithm (EB64) against the other symmetric cryptography techniques such as DES and AES algorithm. Considering and analysing the

experimental results from the graph (figure I and figure II respectively) on the time taken for both encryption and decryption on various sizes of SMS by the three algorithms. It was clearly shown that EB64 algorithm consumes least encryption and decryption time as compared with DES and AES while DES consumes longest encryption and decryption time as equally compared with AES and EB64. In addition, from the comparison encryption time graph, the encryption time of EB64 algorithm is far lesser than both DES and AES algorithm.

Therefore, the graph result of the EB64 algorithm in this research shows that there is an increase in the time taken for both encryption and decryption as the sizes of the messages getting larger.

VI. Contributions to Knowledge

The research work carried out performance evaluation on DES, AES and EB64 respectively and also observed that the encryption and decryption time of EB64 were faster on SMS as compared to AES and DES.

VII. Future work

The future work will be focused on experiments on image and audio data and more focus will be to improve encryption time and decryption time.

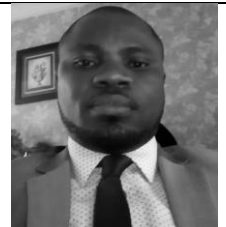
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